GSE223077 - Bulk RNA-seq of ADGRG6 Knocked Down iAT2 Cells Following Cigarette Smoke Exposure[1]

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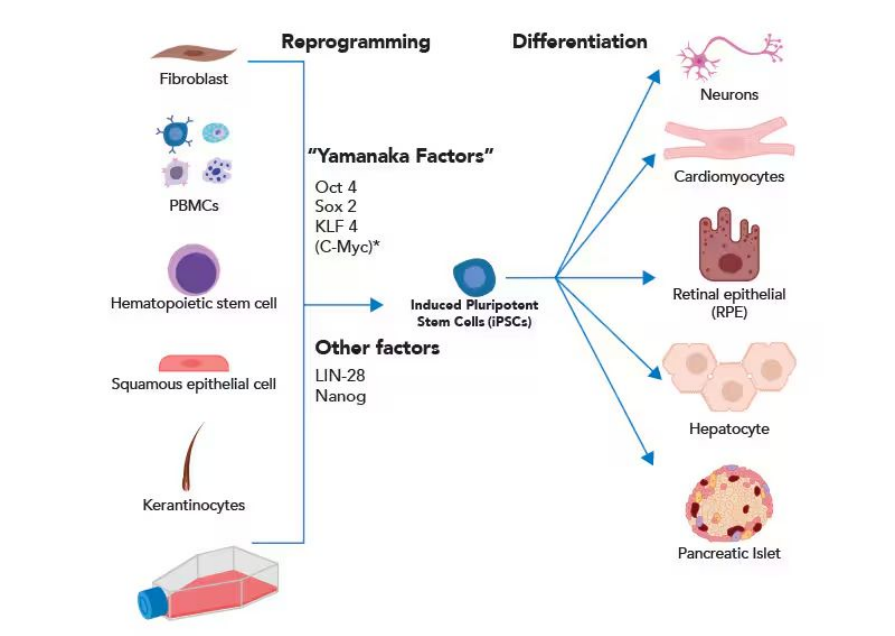
Inducible CRISPRi iPSC Platforms[1]



Full name is inducible CRISPR interference (CRISPRi) human induced pluripotent stem cell (iPSC) platform.

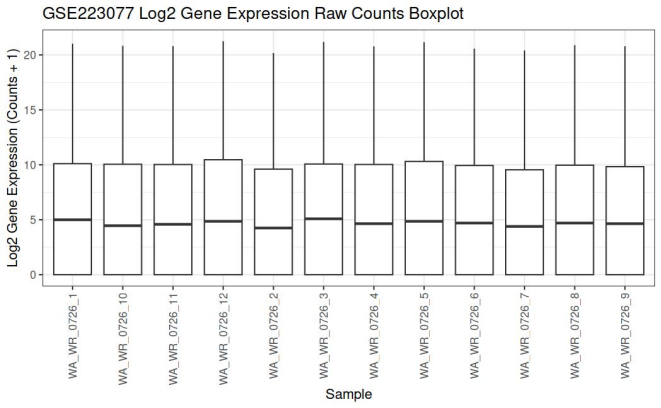
- **Yamanaka Factors**: four transcription factors used to 'reprogram' adult cells back into induced Pluripotent Stem Cells (iPSCs)
- **CRISPRi**: gene-editing technology used to silence a specific gene
  - In this experiment, ADGRG6 gene was knocked down.
- Stem Cell → Type 2 Alveolar Epithelial Cells (AT2 Cells)

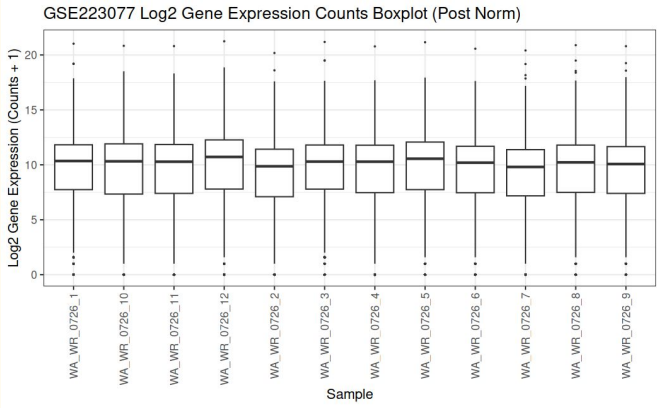
GSE223077 Dataset[1]

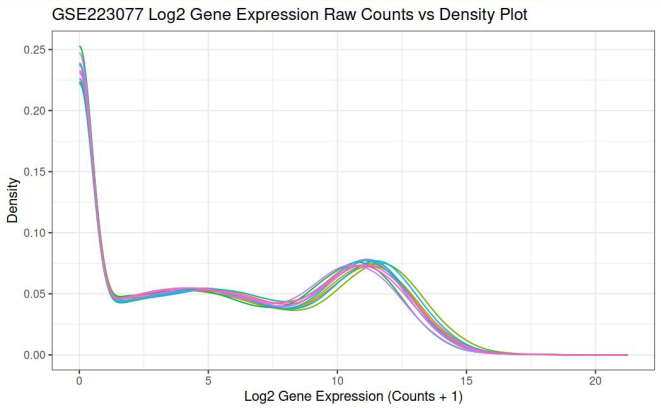
There are 12 Samples named WA_WR_0726_1-12 from this dataset.

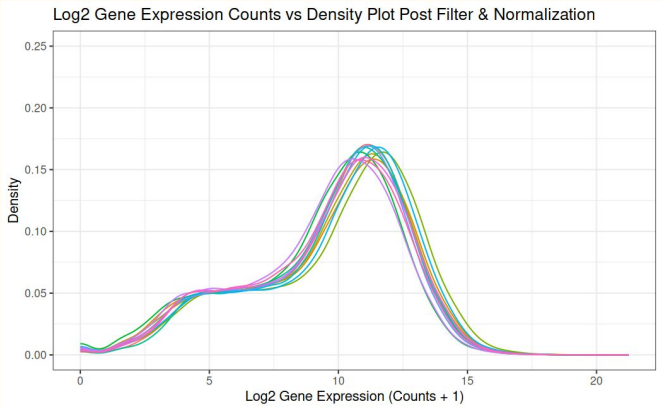
Sample Name	Genotype	Exposure
WA_WR_0726_1	Wild Type(WT)	Air
WA_WR_0726_2	WT	Air
WA_WR_0726_3	WT	Air
WA_WR_0726_4	ADGRG6_kd	Air
WA_WR_0726_5	ADGRG6_kd	Air
WA_WR_0726_6	ADGRG6_kd	Air
WA_WR_0726_7	WT	Smoke
WA_WR_0726_8	WT	Smoke
WA_WR_0726_9	WT	Smoke
WA_WR_0726_10	ADGRG6_kd	Smoke
WA_WR_0726_11	ADGRG6_kd	Smoke
WA_WR_0726_12	ADGRG6_kd	Smoke

Filter by Expression → TMM Normalization(edgeR^2)

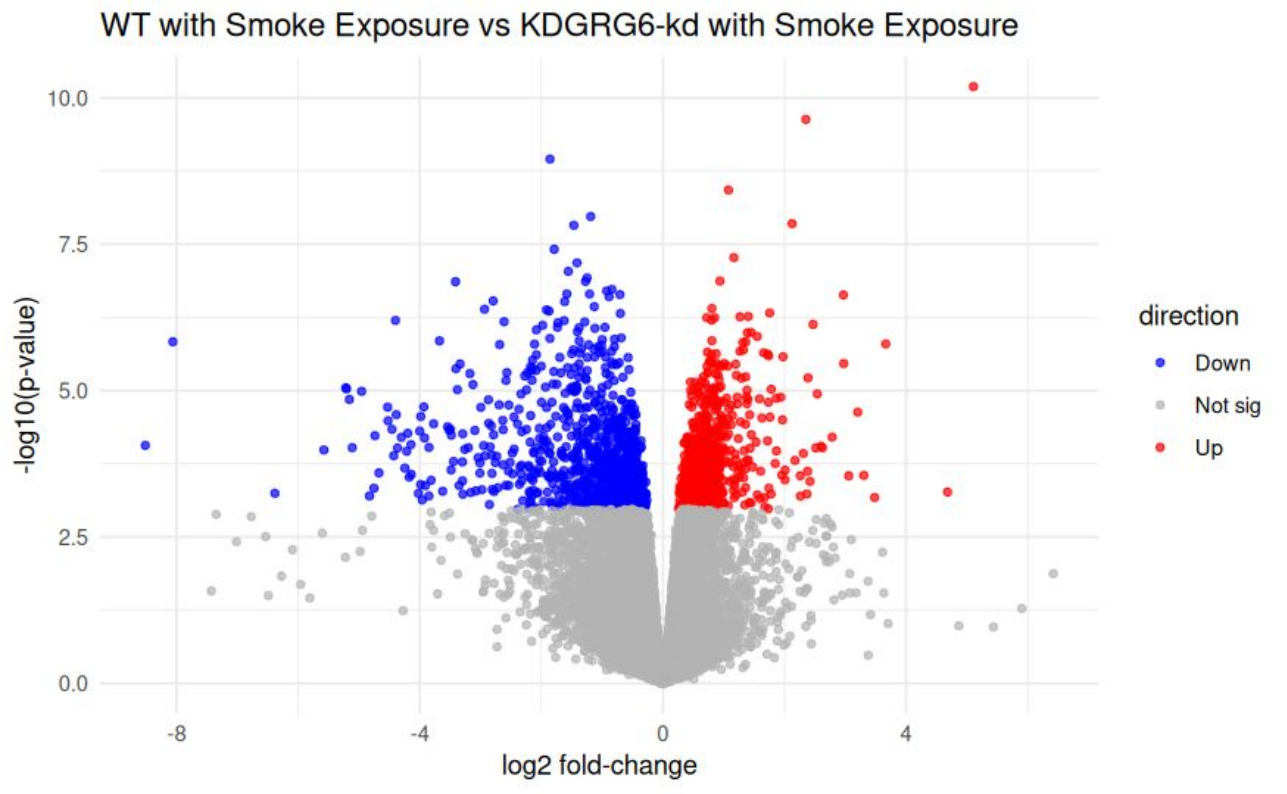








Volcano Plot (WT Smoke vs ADGRD6-kd Smoke)



Top Deferentially Expressed Genes

hgnc_symbol	logFC	logCPM	F	PValue	FDR
NPPC	5.1067181	3.4534993	388.6875664	0	1.00E-06
PPP1R12C	2.3505354	4.3798665	309.9016443	0	1.80E-06
SPON2	-1.8568747	6.2506938	240.2676464	0	5.60E-06
STIM2	1.0790886	6.6782062	196.6854195	0	1.41E-05
FBN2	-1.1890136	5.8393885	165.3622285	0	3.21E-05
RGS16	2.1238792	4.4414665	157.7975669	0	3.24E-05
FN1	-1.4638042	13.0950936	156.061835	0	3.24E-05
LRRN1	-1.7872783	6.0175243	133.1052167	0	7.27E-05
RARG	1.1667422	3.5165704	125.9000568	0.0000001	8.96E-05
GALNT5	-1.4126366	6.118203	121.5404948	0.0000001	9.91E-05

Thresholded Over-Representation Analysis(ORA)[4]

g:Profiler

FDR < 0.01;

$5 <$ Pathway Geneset Size < 500

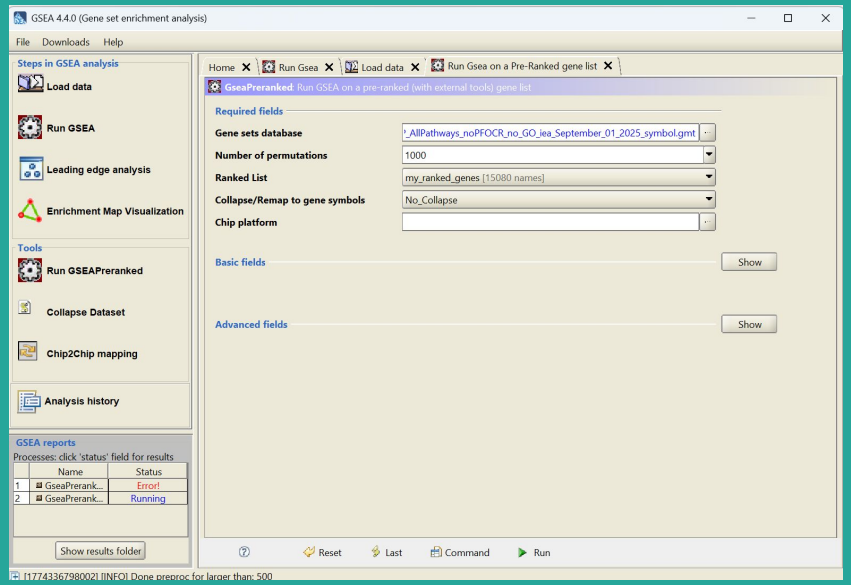
Annotation Database

- GO:BP (Ver 2025-03-16)[5]
- KEGG (Ver 2024-01-22)[6]
- Reactome (Ver 2025-05-23)[7]
- WikiPathways (Ver 2026-03-10)[8]

Top enriched pathways for up-regulated significant genes							
term_name	source	p_value	term_size	query_size	intersection_size	precision	recall
ribonucleoprotein complex biogenesis	GO:BP	0.00E+00	402	6573	334	0.0508139	0.8308458
ribosome biogenesis	GO:BP	0.00E+00	261	6573	227	0.0345352	0.8697318
rRNA processing	GO:BP	0.00E+00	168	6573	147	0.0223642	0.875
rRNA metabolic process	GO:BP	0.00E+00	207	6573	171	0.0260155	0.826087
mRNA processing	GO:BP	0.00E+00	360	6573	257	0.0390993	0.7138889
Ribosome	KEGG	0.00E+00	153	3454	123	0.0356109	0.8039216
Parkinson disease	KEGG	0.00E+00	265	3454	177	0.0512449	0.6679245
Amyotrophic lateral sclerosis	KEGG	0.00E+00	363	3454	223	0.0645628	0.6143251
Nucleocytoplasmic transport	KEGG	0.00E+00	108	3454	79	0.022872	0.7314815
DNA replication	KEGG	0.00E+00	36	3454	34	0.0098437	0.9444444
Translation	REAC	0.00E+00	292	4675	248	0.0530481	0.8493151
rRNA processing in the nucleus and cytosol	REAC	0.00E+00	192	4675	175	0.0374332	0.9114583
rRNA processing	REAC	0.00E+00	202	4675	181	0.0387166	0.8960396
Major pathway of rRNA processing in the nucleolus and cytosol	REAC	0.00E+00	182	4675	166	0.035508	0.9120879
Processing of Capped Intron-Containing Pre-mRNA	REAC	0.00E+00	276	4675	222	0.0474866	0.8043478
Cytoplasmic ribosomal proteins	WP	0.00E+00	88	3748	86	0.0229456	0.9772727
mRNA processing	WP	0.00E+00	126	3748	97	0.0258805	0.7698413
Retinoblastoma gene in cancer	WP	0.00E+00	89	3748	72	0.0192102	0.8089888
DNA replication	WP	0.00E+00	42	3748	40	0.0106724	0.952381
Electron transport chain OXPHOS system in mitochondria	WP	0.00E+00	104	3748	79	0.0210779	0.7596154

Top enriched pathways for down-regulated significant genes

term_name	source	p_value	term_size	query_size	intersection_size	precision	recall
plasma membrane bounded cell projection morphogenesis	GO:BP	0.00E+00	458	5916	258	0.0436105	0.5633188
neuron projection morphogenesis	GO:BP	0.00E+00	443	5916	251	0.0424273	0.5665914
cell projection morphogenesis	GO:BP	0.00E+00	462	5916	259	0.0437796	0.5606061
small GTPase-mediated signal transduction	GO:BP	0.00E+00	391	5916	223	0.0376944	0.5703325
regulation of cell projection organization	GO:BP	0.00E+00	467	5916	256	0.0432725	0.5481799
Axon guidance	KEGG	0.00E+00	181	2908	112	0.0385144	0.6187845
Focal adhesion	KEGG	0.00E+00	202	2908	122	0.0419532	0.6039604
Human papillomavirus infection	KEGG	1.00E-07	330	2908	165	0.05674	0.5
Adherens junction	KEGG	2.00E-07	93	2908	60	0.0206327	0.6451613
Regulation of actin cytoskeleton	KEGG	1.10E-06	227	2908	118	0.0405777	0.5198238
RHO GTPase cycle	REAC	0.00E+00	449	3870	242	0.0625323	0.5389755
Extracellular matrix organization	REAC	0.00E+00	298	3870	172	0.0444444	0.5771812
RAC1 GTPase cycle	REAC	0.00E+00	184	3870	111	0.0286822	0.6032609
CDC42 GTPase cycle	REAC	0.00E+00	155	3870	94	0.0242894	0.6064516
Non-integrin membrane-ECM interactions	REAC	2.00E-07	59	3870	44	0.0113695	0.7457627
Focal adhesion	WP	0.00E+00	198	3117	116	0.0372153	0.5858586
Ciliopathies	WP	4.50E-06	184	3117	103	0.0330446	0.5597826
Epithelial to mesenchymal transition in colorectal cancer	WP	4.70E-06	161	3117	92	0.0295156	0.5714286
Canonical and non canonical TGF B signaling	WP	5.20E-06	17	3117	17	0.005454	1
Primary focal segmental glomerulosclerosis FSGS	WP	4.74E-05	72	3117	47	0.0150786	0.6527778



Bader's Lab[10]:

Human_GOBP_AllPathways_noPFOCR_no_GO_iea_September_01_2025_symbol.gmt

GSEA Report for Dataset my_ranked_genes

Enrichment in phenotype: na

- 2538 / 6155 gene sets are upregulated in phenotype **na_pos**
- 1114 gene sets are significant at FDR < 25%
- 712 gene sets are significantly enriched at nominal pvalue < 1%
- 965 gene sets are significantly enriched at nominal pvalue < 5%
- Snapshot of enrichment results
- Detailed enrichment results in html format
- Detailed enrichment results in TSV format (tab delimited text)
- Guide to interpret results

Enrichment in phenotype: na

- 3617 / 6155 gene sets are upregulated in phenotype **na_neg**
- 918 gene sets are significantly enriched at FDR < 25%
- 440 gene sets are significantly enriched at nominal pvalue < 1%
- 898 gene sets are significantly enriched at nominal pvalue < 5%
- Snapshot of enrichment results
- Detailed enrichment results in html format
- Detailed enrichment results in TSV format (tab delimited text)
- Guide to interpret results

Dataset details

- The dataset has 15080 features (genes)
- No probe set => gene symbol collapsing was requested, so all 15080 features were used

Gene set details

- Gene set size filters (min=15, max=500) resulted in filtering out 12816 / 18971 gene sets
- The remaining 6155 gene sets were used in the analysis
- List of gene sets used and their sizes (restricted to features in the specified dataset)

Gene markers for the na_pos *versus* na_neg comparison

- The dataset has 15080 features (genes)
- Detailed rank ordered gene list for all features in the dataset

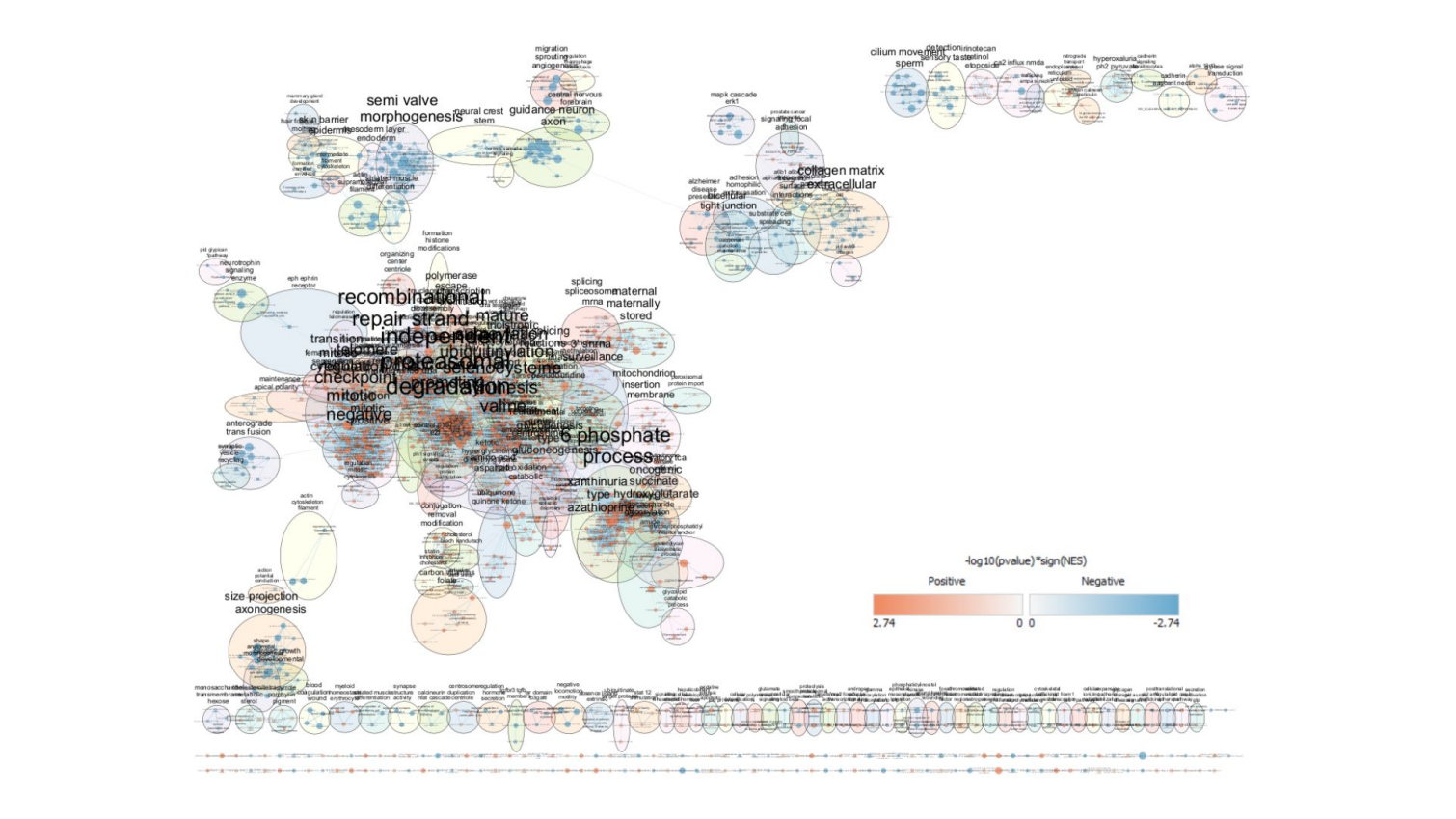
GSEA - Up-regulated Pathways

	GS follow link to MSigDB	GS DETAILS	SIZE	ES	NES	NOM p-val	FDR q-val	FWER p-val	RANK AT MAX	LEADING EDGE
1	HALLMARK_E2F_TARGETS%MSIGDBHALLMARK%HALLMARK_E2F_TARGETS	Details ...	160	0.75	3.23	0.000	0.000	0.000	2670	tags=79%, list=18%, signal=95%
2	PROTEIN SYNTHESIS: TYROSINE%PATHWHIZ%PW120527	Details ...	78	0.80	3.11	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
3	PROTEIN SYNTHESIS: PHENYLALANINE%PATHWHIZ%PW112934	Details ...	80	0.80	3.11	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=109%
4	PROTEIN SYNTHESIS: ASPARTIC ACID%SMPDB%SMP0111858	Details ...	78	0.80	3.09	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
5	PROTEIN SYNTHESIS: GLUTAMINE%SMPDB%SMP0111862	Details ...	78	0.80	3.09	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
6	PROTEIN SYNTHESIS: ISOLEUCINE%SMPDB%SMP0111872	Details ...	78	0.80	3.08	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
7	PROTEIN SYNTHESIS: LEUCINE%SMPDB%SMP0111873	Details ...	78	0.80	3.08	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
8	PROTEIN SYNTHESIS: METHIONINE%PATHWHIZ%PW112933	Details ...	78	0.80	3.08	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
9	PROTEIN SYNTHESIS: GLYCINE%PATHWHIZ%PW112928	Details ...	78	0.80	3.07	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
10	PROTEIN SYNTHESIS: LYSINE%SMPDB%SMP0111874	Details ...	78	0.80	3.07	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
11	PROTEIN SYNTHESIS: HISTIDINE%PATHWHIZ%PW112929	Details ...	78	0.80	3.07	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
12	PROTEIN SYNTHESIS: PROLINE%PATHWHIZ%PW113695	Details ...	78	0.80	3.06	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
13	HALLMARK_MYC_TARGETS_V1%MSIGDBHALLMARK%HALLMARK_MYC_TARGETS_V1	Details ...	148	0.72	3.06	0.000	0.000	0.000	3056	tags=76%, list=20%, signal=94%
14	PROTEIN SYNTHESIS: SERINE%PATHWHIZ%PW120517	Details ...	78	0.80	3.05	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
15	EUKARYOTIC TRANSLATION TERMINATION%REACTOME%R-HSA-72764.6	Details ...	89	0.77	3.05	0.000	0.000	0.000	2478	tags=87%, list=16%, signal=103%
16	HALLMARK_G2M_CHECKPOINT%MSIGDBHALLMARK%HALLMARK_G2M_CHECKPOINT	Details ...	139	0.72	3.05	0.000	0.000	0.000	2943	tags=72%, list=20%, signal=89%
17	SRP-DEPENDENT COTRANSLATIONAL PROTEIN TARGETING TO MEMBRANE%REACTOME%R-HSA-1799339.4	Details ...	109	0.75	3.05	0.000	0.000	0.000	2966	tags=88%, list=20%, signal=109%
18	PROTEIN SYNTHESIS: ALANINE%PATHWHIZ%PW101384	Details ...	78	0.80	3.05	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%
19	TRANSLATION%REACTOME%R-HSA-72766.6	Details ...	272	0.66	3.05	0.000	0.000	0.000	3709	tags=72%, list=25%, signal=93%
20	PROTEIN SYNTHESIS: CYSTEINE%PATHWHIZ%PW112918	Details ...	78	0.80	3.05	0.000	0.000	0.000	2478	tags=91%, list=16%, signal=108%

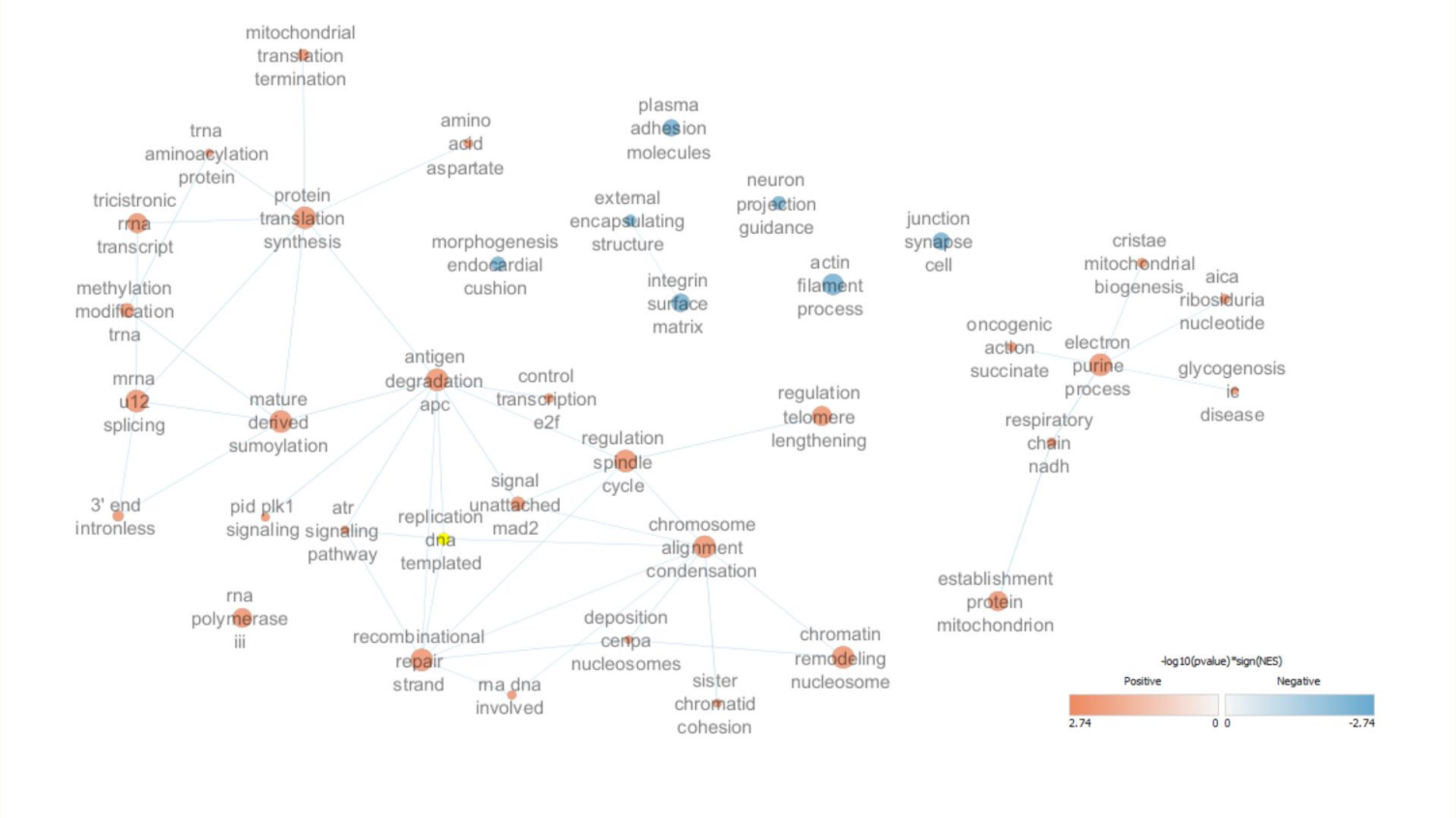
GSEA - Down-regulated Pathways

	GS follow link to MSigDB	GS DETAILS	SIZE	ES	NES	NOM p-val	FDR q-val	FWER p-val	RANK AT MAX	LEADING EDGE
1	CADHERIN SIGNALING PATHWAY%PANTHER PATHWAY%P00012	Details ...	118	-0.58	-2.35	0.000	0.000	0.000	3755	tags=66%, list=25%, signal=87%
2	SKIN DEVELOPMENT%GOBP%GO:0043588	Details ...	85	-0.61	-2.34	0.000	0.000	0.000	2856	tags=49%, list=19%, signal=61%
3	INTERMEDIATE FILAMENT-BASED PROCESS%GOBP%GO:0045103	Details ...	44	-0.68	-2.33	0.000	0.000	0.001	2751	tags=61%, list=18%, signal=75%
4	CELL JUNCTION ASSEMBLY%GOBP%GO:0034329	Details ...	197	-0.54	-2.31	0.000	0.000	0.001	2307	tags=39%, list=15%, signal=46%
5	INTERMEDIATE FILAMENT CYTOSKELETON ORGANIZATION%GOBP%GO:0045104	Details ...	43	-0.68	-2.30	0.000	0.000	0.001	2751	tags=60%, list=18%, signal=74%
6	REGULATION OF AXONOGENESIS%GOBP%GO:0050770	Details ...	67	-0.62	-2.30	0.000	0.000	0.001	3049	tags=57%, list=20%, signal=71%
7	CELL-CELL ADHESION%GOBP%GO:0098609	Details ...	313	-0.50	-2.28	0.000	0.000	0.002	2861	tags=41%, list=19%, signal=49%
8	KERATINIZATION%REACTOME%R-HSA-6805567.5	Details ...	43	-0.67	-2.26	0.000	0.000	0.003	2751	tags=60%, list=18%, signal=74%
9	CELL JUNCTION ORGANIZATION%GOBP%GO:0034330	Details ...	323	-0.49	-2.24	0.000	0.001	0.005	2861	tags=40%, list=19%, signal=49%
10	EPIDERMIS DEVELOPMENT%GOBP%GO:0008544	Details ...	124	-0.54	-2.22	0.000	0.001	0.006	2882	tags=43%, list=19%, signal=52%
11	CELL-CELL JUNCTION ORGANIZATION%GOBP%GO:0045216	Details ...	122	-0.54	-2.20	0.000	0.001	0.011	2398	tags=39%, list=16%, signal=45%
12	ACTIVATION OF MATRIX METALLOPROTEINASES%REACTOME DATABASE ID RELEASE 93%1592389	Details ...	21	-0.75	-2.20	0.000	0.001	0.011	1394	tags=48%, list=9%, signal=52%
13	NEURON PROJECTION GUIDANCE%GOBP%GO:0097485	Details ...	133	-0.53	-2.19	0.000	0.001	0.012	2639	tags=49%, list=18%, signal=59%
14	EXTRACELLULAR MATRIX ORGANIZATION%REACTOME%R-HSA-1474244.5	Details ...	255	-0.49	-2.19	0.000	0.001	0.012	2914	tags=43%, list=19%, signal=52%
15	COLLAGEN DEGRADATION%REACTOME DATABASE ID RELEASE 93%1442490	Details ...	30	-0.68	-2.18	0.000	0.001	0.016	2277	tags=43%, list=15%, signal=51%
16	CELL-CELL ADHESION VIA PLASMA-MEMBRANE ADHESION MOLECULES%GOBP%GO:0098742	Details ...	125	-0.53	-2.17	0.000	0.001	0.017	2386	tags=42%, list=16%, signal=50%
17	GLYPICAN 1 NETWORK%PATHWAY INTERACTION DATABASE NCI-NATURE CURATED DATA%GLYPICAN 1 NETWORK	Details ...	22	-0.74	-2.16	0.000	0.001	0.019	3027	tags=77%, list=20%, signal=97%
18	KERATINOCYTE DIFFERENTIATION%GOBP%GO:0030216	Details ...	37	-0.65	-2.16	0.000	0.001	0.019	2684	tags=49%, list=18%, signal=59%
19	INTERMEDIATE FILAMENT ORGANIZATION%GOBP%GO:0045109	Details ...	30	-0.68	-2.15	0.000	0.001	0.025	2751	tags=60%, list=18%, signal=73%
20	ACTIN CYTOSKELETON ORGANIZATION%GOBP%GO:0030036	Details ...	413	-0.46	-2.15	0.000	0.001	0.025	3695	tags=43%, list=25%, signal=55%

Cytoscape with Enrichment Map Pipline(1.1.0)[11,12]- Raw



Cytoscape[11,12] - Summary Map



Conclusion - Key Take-away

- ADGRG6 (GPR126) is a critical regulator in type 2 alveolar epithelial cells, controlling cell proliferation, cytoskeletal organization, and cell junction integrity.
- Knockdown of ADGRG6 leads to increased cell proliferation but disrupted cytoskeleton and weakened cell-cell adhesion, highlighting its role in maintaining alveolar structure.
- Cigarette smoke exposure amplifies these effects, further promoting proliferation and impairing apical junctions, demonstrating a gene-environment interaction in COPD pathogenesis.
- These findings provide molecular insight into why smoking damages lung structure and underscore the potential of ADGRG6/GPR126 as a therapeutic target for lung repair and COPD intervention.

References

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